

Existing Literature: The following sources are existing research in the field I will be studying. They offer helpful benchmarks on how to work with much of the data I am drawing from, although they do not incorporate macroeconomic and consumer spending variables, nor do they really monitor how the contribution of each variable has changed over the last few decades. These are some of the gaps I hope to address in my project.

1. Elberse, Anita. "The Power of Stars: Do Star Actors Drive the Success of Movies?" *Journal of Marketing* 71, no. 4 (2007): 102–120. <https://doi.org/10.1509/jmkg.71.4.102>.

Using an event-study design built on over 1,200 real-time casting announcements traded on a Hollywood stock-market simulation, Elberse estimates that star involvement adds roughly \$3 million in expected theatrical revenue per film. This approach offers a credible external benchmark for the size of any talent-effect coefficient. It can be referenced as a point of comparison for director/actor-quality coefficients, rather than as a source of ready-made variables.

2. Lash, Michael T., and Kang Zhao. "Early Predictions of Movie Success: The Who, What, and When of Profitability." arXiv preprint, arXiv:1506.05382 (2015). <https://arxiv.org/abs/1506.05382>.

This paper proposes a machine-learning system that separates predictors into “who” (cast/director quality, including a director-profit track record), “what” (genre and plot content), and “when” (release timing and prior-year industry-wide profitability). Regarding my research, it is most useful for its feature-engineering ideas around director quality rather than its predictive methodology.

3. Terry, Neil, Michael Butler, and De’Arno De’Armond. "The Determinants of Domestic Box Office Performance in the Motion Picture Industry." *Southwestern Economic Review* 32, no. 1 (2005): 137–148.

This is a widely cited early study (505 films, 2001–2003) that models box office as a function of critic reviews, award nominations, sequel status, MPAA rating, budget, and release breadth, testing linear and semi-log specifications side by side. This research is most helpful as a historical baseline: a documented snapshot of which film-specific factors mattered in the early 2000s, against which later estimates can be compared to gauge how the determinants have shifted over time.

4. Wang, Siyi. "An Empirical Analysis of Determinants Driving Hollywood Box Office Success." In *Proceedings of the 2025 10th International Conference on Financial Innovation and Economic Development (ICFIED 2025)*, edited by M. Md Husin et al., 398–404. Advances in Economics, Business and Management Research 328. Atlantis Press, 2025. https://doi.org/10.2991/978-94-6463-702-1_43.

This paper conducts a multiple linear regression on roughly 7,400 films that includes a director-reputation variable built from the average historical rating of a director’s prior work, which is the closest existing analogue to an IMDb-based director variable. Wang

concludes that director reputation, log budget, audience rating, and runtime all significantly predict revenue.

Data Sources:

Film and Industry Data:

1. IMDb.com, Inc. *IMDb Non-Commercial Datasets*. Accessed September 15, 2026. <https://data.imdb.com/non-commercial-datasets/>.

These files supply the crew, cast, and rating tables needed to construct a director-quality measure absent from the OpusData extract, matched to it by title and release year. Each director's average rating must be restricted to titles released before the film under study, since including later work would bias the historical measure with information unavailable at the time of release.

2. Nash Information Services, LLC. *OpusData Movie Data Extract*. Accessed September 15, 2026. http://www.opusdata.com/documentation/index.php/Database_Extracts.

This extract constitutes the primary analytical panel, comprising 2,226 theatrical releases spanning production years 2010 through 2025. Each record supplies domestic and international box office totals, production budget, MPAA rating, genre, creative type, source material, production method, sequel status, and running time, together forming the dependent variable and most of the film-specific control variables in the proposed regression.

Macroeconomic and Consumer:

- U.S. Bureau of Labor Statistics. "Average Hourly Earnings of All Employees, Total Private" [CES0500000003]. *FRED*, Federal Reserve Bank of St. Louis. Accessed September 15, 2026. <https://fred.stlouisfed.org/series/CES0500000003>.

Average hourly earnings across the private sector proxy real wage growth, indicating the discretionary income available for entertainment spending near each release date.

- U.S. Bureau of Labor Statistics. "Consumer Price Index for All Urban Consumers: Admission to Movies, Theaters, and Concerts in U.S. City Average" [CUSR0000SS62031]. *FRED*, Federal Reserve Bank of St. Louis. Accessed September 15, 2026. <https://fred.stlouisfed.org/series/CUSR0000SS62031>.

This Consumer Price Index subcomponent tracks admission prices to movies, theaters, and concerts jointly, serving as the closest available federal proxy for national ticket-price trends. Because it blends film admissions with theater and concert admissions, its use as a movie-specific measure carries some imprecision worth acknowledging in the methods section.

U.S. Bureau of Labor Statistics. “Consumer Price Index for All Urban Consumers: Rent of Primary Residence in U.S. City Average” [CUSR0000SEHA]. *FRED*, Federal Reserve Bank of St. Louis. Accessed September 15, 2026.
<https://fred.stlouisfed.org/series/CUSR0000SEHA>.

Rent of primary residence, drawn from the Consumer Price Index, represents a fixed household cost that may compete with discretionary spending on films. Its seasonally adjusted, monthly frequency allows direct alignment with release dates across the sample.

U.S. Bureau of Labor Statistics. “Unemployment Rate” [UNRATE]. *FRED*, Federal Reserve Bank of St. Louis. Accessed September 15, 2026.
<https://fred.stlouisfed.org/series/UNRATE>.

The national unemployment rate operationalizes labor-market slack and directly tests the unemployment-decline component. Monthly, seasonally adjusted values align cleanly with film release dates throughout the panel.

U.S. Energy Information Administration. “US Regular All Formulations Gas Price” [GASREGW]. *FRED*, Federal Reserve Bank of St. Louis. Accessed September 15, 2026.
<https://fred.stlouisfed.org/series/GASREGW>.

Weekly retail gasoline prices proxy a broader category of discretionary transportation costs capable of crowding out leisure spending. Aggregating the series to a monthly average would align its frequency with the other economic controls in the panel.

University of Michigan. “University of Michigan: Consumer Sentiment” [UMCSENT]. *FRED*, Federal Reserve Bank of St. Louis. Accessed September 15, 2026.
<https://fred.stlouisfed.org/series/UMCSENT>.

This index of household expectations regarding personal finances and business conditions forms the principal macroeconomic covariate, merged into the panel by release month to separate sentiment-driven box office fluctuations from film-specific ones. Its monthly coverage extends across the entire sample period without gaps.